Supplemental Material for A Theory-Inspired Framework for Few-Shot Cross-Modal Sketch Person Re-Identification

Table 1: Generalization of Alignment Augmentation on Market1501 (Single-Modality Setting)

Methods	Rank-1 (%)	mAP (%)
IDE (Zheng, Zheng, and Yang 2017)	88.8	71.6
IDE + Alignment Augmentation	**90.0**	**74.9**

Detailed Implementation of the Proposed Method

The process of sketch transformation is a technique that converts color or grayscale images into hand-drawn sketch-style representations. Its core steps include: (1) converting the input image to grayscale; (2) inverting the grayscale image to generate a negative image; (3) applying Gaussian blur to the negative image to simulate shadow effects in sketches; and (4) combining the original grayscale image with the blurred image through a color dodge operation to produce the final sketch effect.

Algorithm 1: Sketch Transformation

Input: Input image I, Gaussian blur kernel size k, Gaussian blur standard deviation σ.
Output: Sketch image I^*.

$I_{\text{gray}} \leftarrow \text{Grayscale}(I)$
$I_{\text{inv}} \leftarrow 255 - I_{\text{gray}}$
$I_{\text{blur}} \leftarrow \text{GaussianBlur}(I_{\text{inv}}, k, \sigma)$
$I^* \leftarrow \text{Dodge}(I_{\text{gray}}, I_{\text{blur}})$
return I^*

The process of sketch transformation is illustrated in Algorithm 1, and the implementation of local transformation is detailed in Algorithm 2. A visual example of the local sketch injection strategy can be found in Fig. 1, providing an intuitive understanding of how the transformation is applied in practice.

Supplemental Experiments

Here provides the ablation studies, augmentation validations, and hyperparameter sensitivity analyses conducted to support the theoretical claims and clarify the effectiveness of each component in our proposed KTCAA framework.

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Algorithm 2: Local Transformation

Input: Input image I.
Parameter: Image size W and H; Area of image S; Transformation probability p; Area ratio range s_l and s_h; Aspect ratio range r_1 and r_2.
Output: Transformed image I^*.

Let $p_1 = \text{Rand}(0, 1)$.
if $p_1 \geq p$ **then**
  Let $I^* = I$.
  return I^*.
else
  while true **do**
    Let $S_t = \text{Rand}(s_l, s_h) \times S$.
    Let $r_t = \text{Rand}(r_1, r_2)$.
    Let $H_t = \sqrt{S_t \times r_t}$, $W_t = \sqrt{S_t / r_t}$.
    Let $x_t = \text{Rand}(0, W)$, $y_t = \text{Rand}(0, H)$.
    if $x_t + W_t \leq W$ **and** $y_t + H_t \leq H$ **then**
      Let $Position = (x_t, y_t, x_t + W_t, y_t + H_t)$.
      Let $I(Position) = t(Position)$.
      Let $I^* = I$.
      return I^*.
    end if
  end while
end if

Generalization of Alignment Augmentation in Single-Modality Settings

To evaluate the generalization capability of the Alignment Augmentation (AA) module beyond cross-modal tasks, we assessed it on the single-modal Market1501 dataset using the IDE baseline (Zheng, Zheng, and Yang 2017). As shown in Table 1, incorporating AA leads to significant improvements in both Rank-1 accuracy and mAP, indicating that simulating visual diversity through sketch-style perturbations can also enhance feature robustness in standard single-modal Re-ID.

Figure 1: Illustration of modality transformation on the Market-1501 dataset. The top row shows the original RGB images, the middle row presents samples with localized sketch-style transformations applied to randomly selected regions, and the bottom row displays fully sketch-transformed versions. This demonstrates the process of modality injection used in our Alignment Augmentation (AA) module to simulate target modality characteristics and improve feature alignment.

Table 2: Comparison of Different Augmentation Strategies

Method	Rank-1 Accuracy (%)	mAP (%)
Baseline (No Augmentation)	12.1	15.3
Baseline + Global Augmentation	15.7	16.5
Baseline + Alignment Augmentation	18.0	19.9

Comparison of Global vs Local Augmentation

To evaluate the effectiveness of localized modality transformation, we compare Alignment Augmentation (AA) with two baselines: no augmentation and global sketch-style transformation. As shown in Tab. 2, AA outperforms both settings, achieving the highest Rank-1 accuracy and mAP. This demonstrates that localized perturbation not only introduces sufficient modality variation but also preserves more structural consistency than global transformation, leading to improved generalization and robustness.

Hyperparameter Sensitivity Analysis

Tab. 3 summarizes the sensitivity of KTCAA to two key hyperparameters in the Knowledge Transfer Catalyst (KTC) module. On the left, we evaluate the impact of the adversarial perturbation bound ε, which controls the magnitude of input perturbations. As ε increases, both Rank-1 accuracy and mAP steadily improve, but training cost also rises. The best trade-off between performance and efficiency is achieved at $\varepsilon = 8$, which is adopted in our main experiments.

On the right, we analyze the effect of the margin parameter ρ, which defines the minimum enforced separation between positive and negative feature pairs in the triplet loss. The model achieves optimal performance at $\rho = 0.5$, indicating that moderate margin strength best facilitates discriminative feature learning while avoiding over-constraining.

Cross-Dataset Testing Performance

To evaluate generalization in unseen domains, we conduct cross-dataset testing between the PKU-Sketch (PKU) and Market-Sketch-1K (M-S) datasets. Tab.4 shows the results when the model is trained on PKU-Sketch and tested on Market-Sketch-1K, and vice versa. In both cases, our model outperforms existing methods. Specifically, in the PKU-Sketch $\rightarrow$ Market-Sketch-1K test, our model achieves 15.4% Rank-1 accuracy and 12.7% mAP, surpassing SS-reID (Lin et al. 2023) (Rank-1: 11.2%, mAP: 10.6%). Similarly, in the Market-Sketch-1K $\rightarrow$ PKU-Sketch test, our method achieves 38.1% Rank-1 accuracy and 32.6% mAP, significantly outperforming SS-reID (Rank-1: 32.8%, mAP: 29.2%). This indicates that the cross-domain generalization capability of SS-reID is relatively limited when additional attribute labels are unavailable. Notably, for fairness, all competing methods were pre-trained on the single-modal Market1501(Zheng et al. 2015) dataset.

Training Efficiency and Cost

Tab. 5 reports the per-batch training time under different configurations on an Nvidia A40 GPU (48 GB memory)

Table 3: Hyperparameter Sensitivity Analysis of KTCAA. Left: sensitivity of adversarial boundary ε. Right: sensitivity of margin ρ in the KTC module.

(a) Sensitivity of ε (PKU-Sketch)

ε	Rank-1 (%)	mAP (%)	Time (s)
–	80.9	80.2	0.510
2	84.5	84.1	0.534
4	87.1	85.2	0.688
8	87.9	86.4	0.895
16	87.9	86.6	1.420

(b) Sensitivity of Margin ρ (KTC)

Margin ρ	Rank-1 (%)	mAP (%)
0.1	82.3	80.7
0.3	86.1	84.5
0.5	**87.9**	**86.4**
0.7	86.5	85.2
1.0	83.8	81.6

Table 4: Cross-dataset testing performance. The model is trained on the PKU-Sketch (PKU) dataset and tested on the Market-Sketch-1K (M-S) dataset under the single-query setting, and vice versa.

Method	**PKU → M-S**		**M-S → PKU**	
	R-1	mAP	R-1	mAP
DDAG (Ye et al. 2020)	2.2	1.9	3.2	1.4
AGW (Ye et al. 2021)	4.3	2.5	8.4	6.3
DEEN (Zhang and Wang 2023)	5.7	4.1	16.2	13.9
SketchTrans (Chen et al. 2022)	8.3	7.9	20.2	19.4
UNIReID (Chen, Ye, and Jiang 2023)	9.8	9.3	26.4	26.6
SS-reID (Lin et al. 2023)	11.2	10.6	32.8	29.2
Ours	**15.4**	**12.7**	**38.1**	**32.6**

Table 5: Training Time Cost per Batch (Batch Size = 64, $\varepsilon = 8$)

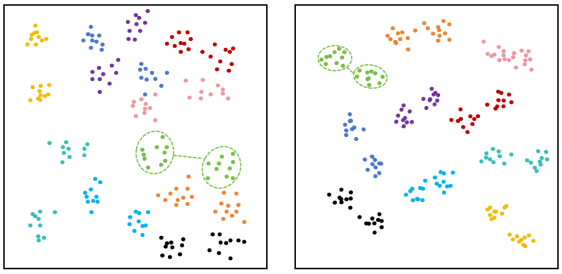
Configuration	Baseline	+ AA	+ AA + KTC
Time Cost (s)	0.503	0.510	0.895

with a batch size of 64 and $\varepsilon = 8$. Introducing the Alignment Augmentation (AA) module causes only negligible overhead, while adding the Knowledge Transfer Catalyst (KTC) increases training time due to adversarial optimization. However, KTC does not affect inference-time efficiency and can be selectively applied during training to balance performance and cost.

As shown in Tab.5, the experiments demonstrate that KTCAA achieves substantial robustness and generalization improvements at only a modest computational cost.

Feature Distribution Visualization

Figure 2 illustrates the t-SNE projection of feature representations from 10 randomly selected identities in the PKU-Sketch dataset. Each identity is visualized using a unique color. The visualization demonstrates that our KTCAA method yields more compact clusters and better inter-class separability than the baseline, confirming the improved alignment of cross-modal features.



(a) Baseline (b) KTCAA

Figure 2: t-SNE visualization of 10 randomly selected identities, where each identity is represented by a unique color. Compared to the baseline, KTCAA produces more compact intra-identity clusters and improved separation across modalities.

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